

SMART INDIA HACKATHON 2026

NETRA — Neural Explainable Triage for Retinal Assessment



SMART INDIA
HACKATHON
2026

Problem Statement ID – 26038

Problem Statement Title – Explainable AI for Diabetic Retinopathy

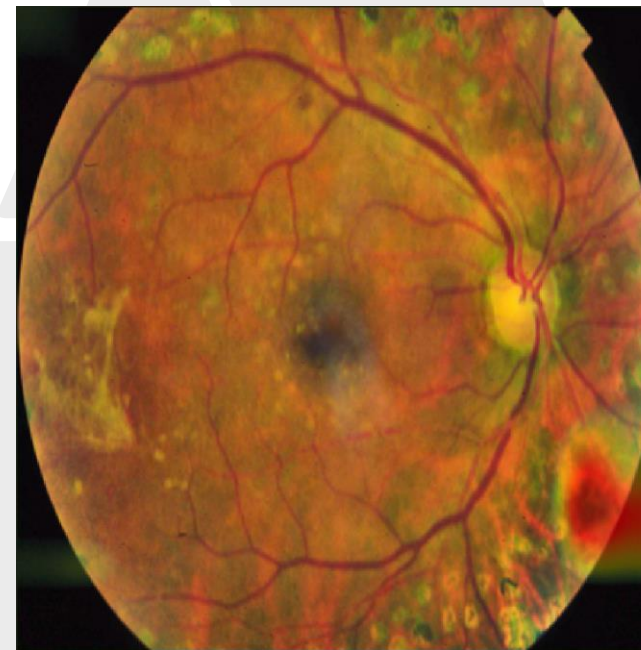
Screening in Rural India

Theme – MedTech / BioTech / HealthTech

PS Category – Software

Team ID – <<TEAM ID>>

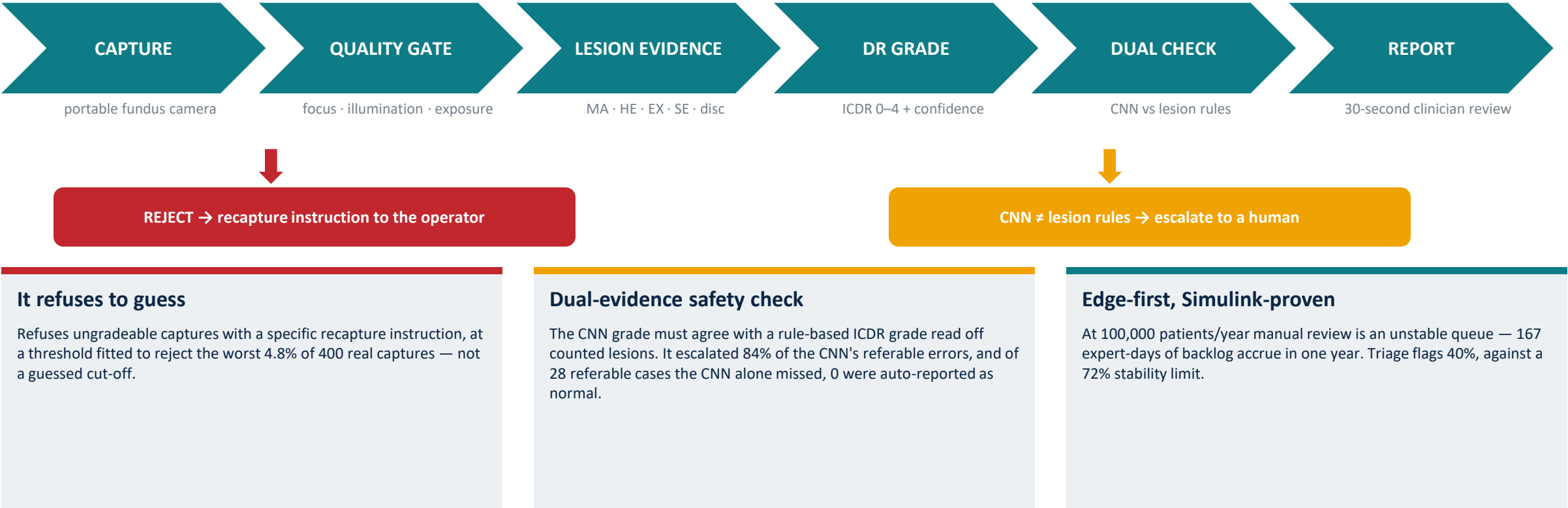
Team Name – <<TEAM NAME>>



Grad-CAM attention on a held-out test image — the evidence behind the grade

IDEA TITLE

❖ *Proposed Solution (Describe your Idea/Solution/Prototype) · Detailed explanation of the proposed solution · How it addresses the problem · Innovation and uniqueness of the solution*



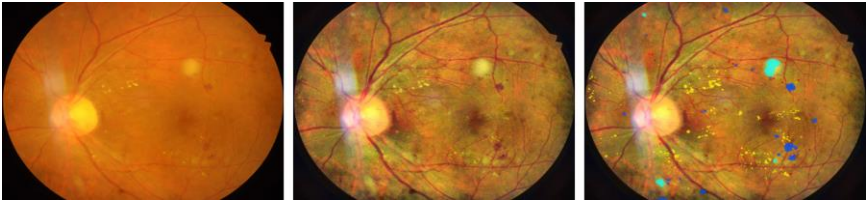
Not a black box — an evidence-based triage system that shows its work and knows when to abstain.

• Technologies to be used (e.g. programming languages, frameworks, hardware) · Methodology and process for implementation (Flow Charts/Images/ working prototype)

M1 QUALITY ASSESSMENT	Image Processing
imgradient · fspecial('laplacian') · adapthisteq · regionprops → score 0–100, thresholds fitted on 400 real captures	
M2 ADAPTIVE ENHANCEMENT	Image Processing
imgaussfilt illumination field · adapthisteq (CLAHE) · imnlmfilt denoise	
M3 LESION SEGMENTATION	Deep Learning + CV
ResNet-18 encoder + U-Net decoder, 5 channels, native resolution, focal-Tversky loss	
M4 GRADING + EXPLAINABILITY	Deep Learning
resnet18 · trainnet · gradCAM · operating point at sensitivity ≥ 0.90	
M5 DEPLOYMENT SIMULATION	Simulink
discrete-time backlog model $B(k+1)=\max(0,B(k)+\text{demand}-\text{capacity})$, two arms on identical arrivals	

- Image Processing
- Computer Vision
- Deep Learning
- Statistics & ML
- Parallel Computing
- Simulink

verified installed, R2026a



raw → enhanced → lesion overlay

Dataset	Images used	Role in this build
APTOS 2019	3,662	ICDR 0–4 grading — train / val / sealed test
IDRiD (Indian)	81 pixel-annotated	Lesion ground truth — 44 train / 10 val / 27 sealed test
Neovascularisation	0	No pixel-level labels exist — not modelled, stated as a limit

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Analysis of the feasibility of the idea
 ·
 Potential challenges and risks
 ·
 Strategies for overcoming these challenges

WHAT IS ACTUALLY BUILT AND MEASURED

Quality gate + enhancement	built+calibrated
Lesion segmentation (5 channels)	trained+evaluated
DR grader (ResNet-18, ICDR 0–4)	trained+evaluated
Grad-CAM explainability	built
Dual-evidence check	built+measured
App Designer GUI	built
Simulink district model	built + simulated (core Simulink)
Neovascularisation detection	not built — no labels exist

Status words, not percentages — a percentage of a module is not a measurable thing.

COST PER SCREENING

₹6–12 per screening, camera amortised	₹0 software licence (academic)	₹500–1,500 private consult avoided
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MATLAB R2026a trial (expires 4 Oct 2026): Image Processing, Computer Vision, Deep Learning, Statistics & ML, Parallel Computing and Simulink, all verified installed. SimEvents is licensed but not installed on this machine, so the district model is built from core Simulink blocks. Trained on one RTX 4060 laptop GPU — lesion net 63 min, grader 11 min.

RISKS AND WHAT ABSORBS THEM

Poor field image quality	quality gate refuses and instructs recapture
Class imbalance (few grade 3–4)	inverse-sqrt class weights + augmentation
Rural bandwidth limits	edge inference, only flagged cases uploaded
Clinician distrust of AI	lesion evidence + Grad-CAM + human-in-the-loop
Domain shift across cameras	illumination normalisation; external validation is future work
54 annotated training images	the IDRiD benchmark's own limit — stated, not hidden

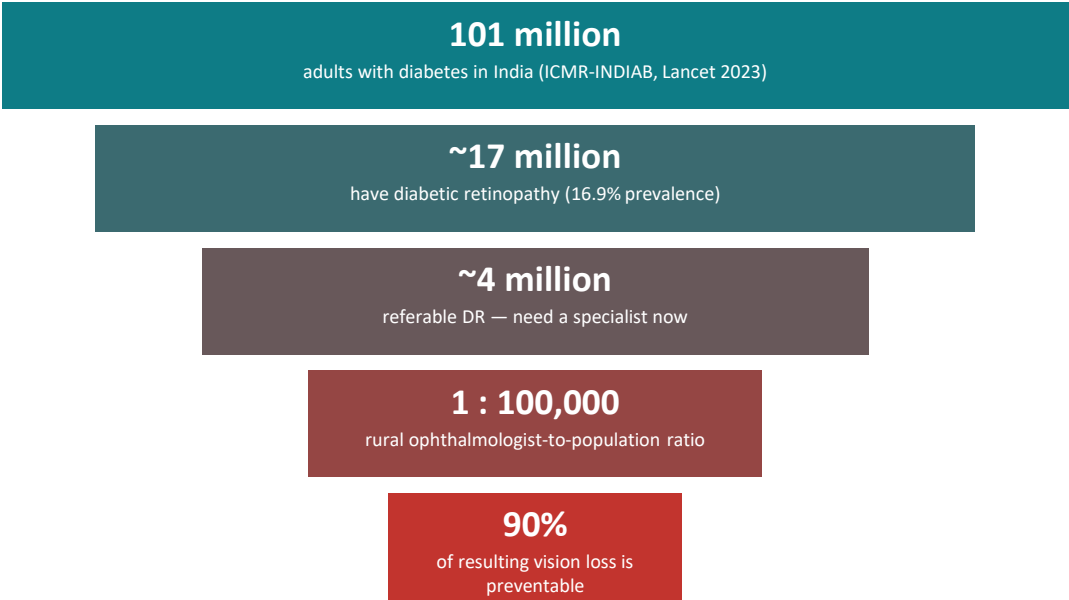
DEPLOYMENT READINESS



Prototype stage is where this build actually is today.

• Potential impact on the target audience · Benefits of the solution (social, economic, environmental, etc.)

WHY SCREENING, NOT TREATMENT, IS THE GAP



The gap is not treatment. The gap is screening capacity.

Social

sight preserved for patients who would never reach a specialist

Economic

averts the lifetime cost of avoidable blindness per case

Systemic

turns one scarce ophthalmologist into district-wide coverage

Aligned

supports Ayushman Bharat HWCs and NPCBVI screening targets

ONE DISTRICT, 100,000 PATIENTS PER YEAR

	✗ Manual	✓ With NETRA
Patients screened / year	100,000	100,000
Images an expert must read	100,000	40,073
Ophthalmologists (FTE)	1.7	0.6
Expert reading time / day	10 hrs of work, 6 available	200 min
Upload volume / day	0.86 GB	0.35 GB
Result turnaround	2–4 weeks	10 s, at the point of care

Manual review is not a staffing shortfall — it is an unstable queue. The Simulink model accumulates 167 expert-days of backlog in one working year. An expert can absorb 72% of captures before the queue diverges; NETRA flags 40%.

districtSim.m + buildNetraSimulink.m, driven by the grader's own held-out triage rate. Assumptions named in the scripts.

• Details / Links of the reference and research work

REFERABLE DR (ICDR 2+) — PUBLISHED SYSTEMS AND THIS BUILD

System	Dataset	Sensitivity	Specificity	Source
Gulshan et al. 2016 (Google)	EyePACS-1	90.3%	98.1%	JAMA 316(22):2402
Gulshan et al. 2016	Messidor-2	96.1%	93.9%	high-sensitivity point
IDx-DR pivotal (Abràmoff 2018)	900 primary-care	87.2%	90.7%	npj Digital Med 1:39
PS 26038 requirement	referable DR	> 90%	> 85%	the problem statement
NETRA (this build)	APTOS held-out, n = 549	91.9%	95.4%	AUC 0.983 · s12_eval_grader

Operating point chosen on the VALIDATION split to meet sensitivity ≥ 0.90 , then applied unchanged to a sealed test split — the protocol Gulshan 2016 and the IDx-DR trial both use. Different dataset from the rows above, so this is a like-for-like task, not a like-for-like benchmark.

LESION-LEVEL RESULTS — 27 SEALED IDRiD TEST IMAGES (Dice)

0.732 Hard exudate	0.590 Haemorrhage	0.546 Soft exudate	0.519 Microaneurysm	85.1% sens Haemorrhage flag (region)
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Datasets

APTOS 2019 — kaggle.com/c/aptos2019-blindness-detection
IDRiD — ieee-dataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid
Both used under their published licences (CC-BY-4.0 for IDRiD)

Clinical standards

Wilkinson et al., International Clinical DR Severity Scale, Ophthalmology 2003;110:1677
ICMR-INDIAB, Lancet Diabetes & Endocrinology 2023
NPCBVI, Ministry of Health & Family Welfare

Methods and tooling

Selvaraju et al., Grad-CAM, ICCV 2017
Porwal et al., IDRiD, Medical Image Analysis 2020;59:101561
MathWorks: gradCAM, adapthisteq, trainnet, Simulink